

etash@cmu.edu
(412) 439-8791

Etash Jhanji
etashj.github.io

linkedin.com/in/etashj
github.com/etashj

Education

Carnegie Mellon University

Bachelor of Science in Computer Science, Concentration in Computer Systems

Pittsburgh, PA

Dec 2028

- **GPA:** 3.9/4.0
- **Courses:** Databases [15-445], Computer Systems [15-213], Imperative Computation [15-122], Theoretical Ideas in CS [15-251], Functional Programming [15-150], Type Theory [98-317], AI/ML [07-280]

Work Experience

Teaching Assistant: Principles of Imperative Computation (15-122)

Carnegie Mellon University School of Computer Science

Pittsburgh, PA

Jan 2026 – Present

- Upgrading legacy server and web applications to Microsoft Entra, Apache and Ubuntu 24.04; replacing shared passwords with fine-grained access prior to onboarding protected records for over 8,500 students
- Containerizing course applications serving 500 students, standardizing deployment and reducing setup time by 50% for leads
- Teach weekly recitations and bootcamps for 50+ students on C/C0, memory management, pointer arithmetic, data structures, and contract-based programming.

CMU Courses Tech Lead

ScottyLabs

Pittsburgh, PA

Sep 2025 – Present

- Migrating course data to client-side DuckDB WASM for $5\times$ faster queries than the original implementation, supporting course search, evaluation aggregation, degree planning, and combined, alternate, and concurrent course dependencies.
- Lead the rewrite of CMU Courses, a product used by 3,000+ students; establish code quality, style, and merge requirements for contributors
- Coordinated with senate, design, and outreach teams to release 5+ years of course data publicly, enabling API access for other apps and agentic workflows while complying with university policy

Research Experience

Undergraduate Researcher

Sunshine Group @ CMU Software and Societal Systems Department

Pittsburgh, PA

Jun 2026 – Present

- Built a tool that elicits formal, property-based specifications from underspecified prompts making requirements explicit before code generation; benchmarked on an 11-task suite derived from found tasks
- Designed a 4-step spec-elicitation pipeline: clarification, elicitation, preview, and generation; developed a suite-coverage score by mutation testing and fuzzing

Projects

- **etash:** Built a POSIX-like shell in Rust featuring a custom modular lexer/tokenizer, word expander, and expression parser with multi-line input state and tests for each component; 10% lower peak memory usage and 40% smaller executable than zsh
- **Futile March:** A complete WebGL game running at 60 fps compressed into a 15KB distribution, loading in $<2s$, with custom GLSL ray marching and procedural physics, packaged as a PWA with offline support, 3D graphics, code-size optimization, and online extensibility such as music and multiplayer
- **SongSorter:** Trained models using PyTorch to extract mood from songs relative to Russell's model of affect, created a Java application to interact with data extracted from Spotify clips
- **AniMatch:** Developed a mobile-responsive "Tinder-style" web application using React with TailwindCSS leveraging the AniList API to provide personalized anime and manga recommendations via a swipe-based interface

Publications

- **Jhanji, E.** Exploring and Applying Audio-Based Sentiment Analysis in Music (2024). *arXiv:2403.17379 [cs.SD]*. doi.org/10.48550/arXiv.2403.17379.
- Maggiore JC, Przepiorski A, Han H, **Jhanji E**, Streeter EC, McDaniels M, Hukriede NA. Loss of Histone Deacetylase 8 Enhances Proximal Tubule Acute Kidney Injury Repair. (2023). Manuscript in preparation.

Skills

Languages: C, C++, Rust, Go, Python, Java, TypeScript, JavaScript, SQL, SML

Technologies: Linux/Unix, Docker, Nix, Git, Vim, PostgreSQL, DuckDB, PyTorch, SvelteKit, TailwindCSS, React

Honors & Awards

- **TartanHacks AppLovin Axon Sponsor Track Winner:** Unique idea category for Futile March in 2026
- **Coding State Champion:** PA Technology Student Association coding event involved a theory test and timed challenges in 2024
- **First Award:** Awarded at the regional and state level by the PA Junior Academy of Science in 2021, 2023, 2024, 2025